SeatNo._____

GANPAT UNIVERSITY
B.Sc(IT) SEM – III REGULAR EXAMINATION

U23A1DFS- DATASTRUCTURE

Regular Nov - Dec 2022 Total Marks: 60

TIME: 3 Hrs.
Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

Q.1 Answer the Following(Any seven) **30**

1 What is Data Structure?Explain it.

2. Explain Double Linklist with Any five Algorithms.

3. Explain Polish Notation (Prefix, Infix, Postfix).

4. Explain Non Linear Data Structure.

5.Explain Explain Height Balanced Tree (AVL).

6. What is stack? Explain the algorithm of push and pop.

7. Explain Insertion Sorting.

8.Explain Graph Traversal Algorithms – BFS .

9.Explain circular Queue With Algorithms .

10.Explain Sequential Searching.

Q.2 Answer the Following(Any one) **05**

1.Write Algorithm for display and find in array

2. Explain Linear Data Structure.

Q.3 Answer the Following(Any two) **10**

1.Explain Simple Queue With Algorithms .

2.Explain Reverse LinkList.

3. Explain Recursion in Linear Data Structure.

Q.4 Answer the Following(Any two) **10**

1. Write a Short note on Binary Tree and its Representation

2 .Explain B Tree and B++ Tree.

3.write a short note on Height Balanced Tree.

Q.5 Answer the Following(Any one) **05**

1.Explain Binary Searching .

2.Explain Selection Sorting.

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GANPAT UNIVERSITY
B.Sc. IT SEM-III EXAMINATION CBCS

U23A2DT1: APPLICATION DEVELOPMENT TOOL-1

Time: 3 Hours] Nov - Dec 2022 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX) (30)

A. What are the various misconceptions of ERP?

B. List out the benefits of ERP to the organization.

C. Discuss different features of Python.

D. How does List differ from Tuple?

E. What is Exceptions? Explain handling an exception condition with suitable example.

F. What is Python Module? Explain creation and use of module with suitable Example.

G. Explain different steps of creating a Database in Odoo.

H. Write a short note on Model Fields.

2. Answer the following: (ANY ONE) (06)

A. Briefly explain about Functional Modules of ERP software?

B. Discuss various challenges of ERP implementation.

3. Answer the following: (ANY ONE) (06)

A. Write a code: "To print Even Numbers in given range of 2 to 40".

B. Explain any three functions of String.

4. **Answer the following: (ANY ONE)** (06)

A. What is the use of Dictionary in python programming? Explain it with suitable example.

B. What is Constructor? Explain how to declare and use constructor in python program.

5. **Answer the following: (ANY ONE)** (06)

A. Discuss about the Directory Structure (Module Structure) of Odoo.

B. Explain any three Structure tools used to design E-Commerce/WEB-SITE.

6. **Answer the following: (ANY ONE)** (06)

A. What is View? Explain Tree/Form/Search Views with Example.

B. Discuss about Model Inheritance in detail.

END OF PAPER

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GANPAT UNIVERSITY

(BSC IT) SEM-III EXAMINATION

U23A3DM2: DATABASE MANAGEMENT SYSTEM - II

Nov - Dec 2022

[Time: 3 Hours] **[Total Marks: 60]**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

Q-1. Answer the following: (Any Six) **(30)**

I. Discuss the features of Oracle 10g.

II. What are the steps to start SQL* plus? What is the use of it?

III. Differentiate between locally managed and Dictionary managed tablespaces.

IV. Discuss the block structure of pl/sql program.

V. What is System Global Area (SGA)? Discuss various parts of SGA.

VI. Explain Enterprise Manager in detail.

VII. Write a note on: Table spaces in Preconfigured databases.

VIII. Discuss User defined exception handling with example.

Q-2. Answer the following: (Any Two) **(08)**

I. Give the difference between SGA and PGA.

II. Discuss SYSTEM and SYSAUX table spaces in short.

III. Write a note on: Oracle Instance Management.

Q-3. Answer the following: (Any one) **(02)**

I. Briefly explain database management framework component.

II. Which commands are used to start and stop the database control?

Q-4. Answer the following: (Any two) **(10)**

I Write down the steps for creating Table space using Enterprise manager.

II Discuss all actions with Table Space.

III Explain the structure of Database Block.

Q-5. Answer the following: (Any Two) **(10)**

I What is cursor? Which are the types of cursors? Explain in detail.

II Discuss the use of procedure with suitable example.

III What is trigger? Explain its types with suitable example.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. IT SEM – III Regular EXAMINATION

U23A4ACN: Advanced Computer Networks

Nov - Dec 2022

TIME: 3 Hrs.] [Total Marks: 60]

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

Q-1. Answer the following Questions. (Any Six) (30)

1. **What is OSI Model? Explain the functions and protocols and services of each layer.**
2. **What are the types of multiplexing? Explain all in brief.**
3. **Explain Data Communication and Data Flow.**
4. **What is Security? Explain Types of Security.**
5. **What is VLAN? Describe its Advantages.**
6. **Explain cryptography attacks.**
7. **What is domain name system? How it works?**
8. **Define Frame tagging.**
9. **Differentiate Circuit Switching and Packet Switching.**
10. **Define Baseband Connection and Broadband Connection.**

Q-2. Answer the following Questions. (Any One) (06)

1. **What is TCP/IP Model? Explain the functions and protocols and services of each layer.**
2. **Explain all transmission media**

Q-3. Answer the following Questions. (Any One) (06)

1. **Difference between Static Routing and Dynamic Routing.**
2. **Differentiate between half duplex and full duplex connection.**

Q-4. Answer the following Questions. (Any Two) (10)

1. **Describe cisco three layer hierarchical model.**
2. **Explain Ethernet and Explain all Types of Ethernet cabling.**
3. **Explain Ethernet Datalink Layer & Ethernet Physical Layer.**

Q-5. Answer the following Questions. (Any Two) (08)

1. **Explain Packet switching method uses two routing methods.**
2. **Explain VLAN Membership with its Methods.**
3. **what is computer Network? Explain with figure.**

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. (IT) SEMESTER - III EXAMINATION (CBCS)

Nov - Dec 2022

U23A5FMI- FUNDAMENTAL MATHEMATICS FOR INFORMATION TECHNOLOGY

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (Any Six)** **(30)**

a. Write a short note on Set Theory

b. If A = {0,1,2,3,4} B = {x / x2 + x – 6 = 0 and x ∈ A} and
C = {x / x2 + x – 12 = 0 and x ∈ A }
Then verify A ∩ (B U C)= (A ∩ B) U (A ∩ C)

c. Prove that .
$n(A \cup B \cup c) = n(A) + n(B) + n(c) - n(A \cap B) - n(B \cap C) - n(A \cap c) + n(A \cap B \cap c)$

d. Prove: $\binom{17}{6} = \binom{16}{5} + \binom{16}{6}$

e. Explain Bipartite graph in detail.

f. Compute: **(1)** $\binom{8}{2}$ **(2)** $\binom{13}{7}$

g. Write a short note on Non-Singular Matrix with an example

h. $If\ A = \begin{bmatrix} 2 & -5 \\ 3 & 1 \end{bmatrix} find\ A^2\ and\ A^3$

2. **Answer the following: (Any Two)** **(10)**

a. In a joint family of 12 persons,7 takes tea , 6 take milk and 2 take neither ,How many members take both tea and milk ?

b. Prove that De Morgan's laws

c. Explain Cartesian Product of Two Sets with an example

3. **Answer the following: (Any One)** **(06)**

a How many ways can the letters can be arranged?
(a) HIPPOPOTAMUS (b) CALCULATOR

b In how many different ways can the letters of the word 'INDIAN' be arranged?

4. **Answer the following: (Any One)** **(06)**

a Define Terms: Multigraphs, Loop, Sub Graphs, Finite graphs, Trivial Graph Complete Graph

b Write a short note on Regular Graph.

5. **Answer the following: (Any Two)** **(08)**

a Define Terms: Matrix, Vector, Square Matrix, Zero Matrix

b $A=\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$

Then find out **1)** AA and **2)** F(A), where $F(x) = x^2 - 2x^2$

c

If $A=\begin{bmatrix} 6 & 4 & -4 \\ 3 & 3 & 7 \end{bmatrix}$ $B=\begin{bmatrix} 2 & 6 & -5 \\ 7 & 4 & 5 \end{bmatrix}$

Then find out **1)** A + B and **2)** A - B

END OF PAPER